Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
12.	*(a)*	(i)		The iodide ions (moved to the anode and) were oxidised / lost electrons forming iodine (1) $2I^-(aq) \rightarrow I_2(aq) + 2e^-$ (1) (Aqueous) iodine was produced giving a yellow/brown coloration (around the anode) (1)	1	1	1	3		1
		(ii)		Both iodide and chloride ions formed precipitate – yellow and white observed (1) On adding ammonia and shaking the white precipitate / silver chloride dissolves (1) Leaving silver iodide as a pale yellow solid / silver iodide does not dissolve (1)		3		3		3
		(iii)		Number of moles of potassium iodate(V) $= \frac{0.100 \times 18.00}{1000}$ $= 1.8 \times 10^{-3}$ (1) Number of moles of NaI present in 25.0 cm^3 of the solution of the mixture $= 1.8 \times 10^{-3} \times 2 = 3.6 \times 10^{-3}$ $\therefore$ Number of moles of sodium iodide in 250 cm^3 = 0.036 (1) Mass of sodium iodide present in 250 cm^3 of the solution of the mixture = 0.036 × 150 = 5.40 g $\therefore$ % sodium iodide in the mixture $= \frac{5.40 \times 100}{11.24} = 48$ (1) ecf possible		3		3	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	*(b)*			$f = c/\lambda = 3 \times 10^8/278 \times 10^{-9} = 1.079 \times 10^{15}$ (Hz) (1) $E = hf \quad \therefore E = 6.63 \times 10^{-34} \times 1.079 \times 10^{15}$ J $= 7.154 \times 10^{-19}$ J (per molecule) $\therefore$ per mole $7.154 \times 10^{-19} \times 6.02 \times 10^{23}$ J mol^{-1} $= 4.307 \times 10^5$ J mol^{-1} (1) $= 431$ kJ mol^{-1}			2	2	2	
	(c)			As the group is descended the bond energies decrease and the wavelengths increase / astatine is below iodine in the Periodic Table / $\lambda_{max} > 400$nm (1) coloured gas linked with the visible region (1)			2	2		
				Question 12 total	**1**	**7**	**5**	**13**	**3**	**4**